23 Maryam is trying to expand and simplify $(n-2)^2$
Here is her working.

$$(n-2)^2 = (n-2)(n-2)$$
$$= n^2 - 2n - 2n - 4$$
$$= n^2 - 4n - 4$$

Maryam's answer is wrong.

(a) Find Maryam's mistake.

1 mark
Sign of final product

(-2) × (-2) = +4, not-4.

Correct expansion:n^2 - 4n + 4.

(1)

Josh is trying to factorise $x^2 - 6x + 8$
His reasoning is,

because $4 \times 2 = 8$
and $4 + 2 = 6$

then $x^2 - 6x + 8 = (x + 4)(x + 2)$

(b) Explain what is wrong with Josh's reasoning.

1 mark
Wrong sum sign

The two numbers must add to-6, not+6.

Use-4 and-2: product+8, sum-6.

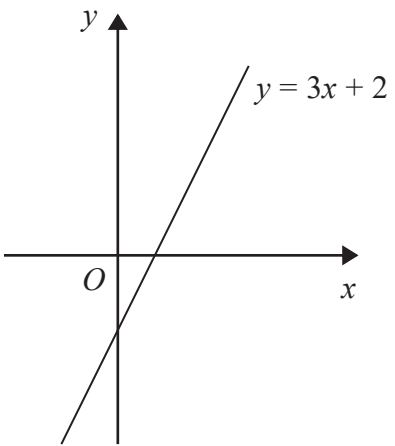
(1)

×	x	-4
x	x^2	$-4x$
-2	$-2x$	8

$x^2 - 6x + 8$

$= (x - 4)(x - 2)$

Shona has to draw the line with equation $y = 3x + 2$
Here is her line.



(c) Explain why Shona's line **cannot** be correct.

1 mark
Wrong intercept sign

At x=0, y=+2, so the y-intercept must be positive.

Her line crosses the y-axis below0: a negative intercept.

(1)

(Total for Question 23 is 3 marks)